

Method Design Document

- **Project Title:** Bank Loan Analysis
- **Domain:** Finance
- **Author:** Ankush Saini (Data Analysis Intern at **SYNTECXHUB**)
- **Tools Used:** Python (Pandas, Scikit-learn, Seaborn), Power BI, SQL
- **Dataset:** financial_loan.csv

2. Project Overview

- **Critical Problem:** The bank faces financial risk due to loan defaults. Identifying the characteristics of high-risk applicants is essential for minimizing financial losses and optimizing lending strategies.
- **Primary Objectives:**
 - *Exploratory Data Analysis (EDA):* Identify key drivers of loan defaults.
 - *Data Cleaning:* Handle missing values and correct data types for robust analysis.
 - *Modelling:* Develop predictive models to assess feature importance and predict total payment outcomes.
 - *Visualization:* Create an interactive Power BI dashboard for real-time risk monitoring.
- **Scope:** The analysis covers a dataset of **38,576** individual loan applications, focusing on geographical regions across the United States, temporal trends throughout the year 2021, and specific numerical constraints like Debt-to-Income (DTI) and interest rates.

3. Dataset Description

- **Source Details:** The primary dataset is financial_loan.csv, containing historical loan application records, customer demographics, and financial indicators.

Summary Table

Metric	Value
Raw Records	38,576
Cleaned Records	38,576
Total Columns (Raw)	24
Total Columns (Engineered)	27

Missing Values	1,438 (in <i>emp_title</i>)
Target Variable Range	Good Loan (86.18%) vs Bad Loan (13.82%)

Column Descriptions Table

Column Name	Data Type	Description
id	Integer	Unique identifier for each loan application
loan_amount	Integer	Total amount of money requested by the borrower
int_rate	Float	Interest rate applied to the loan
installment	Float	Monthly payment owed by the borrower
grade	Object	Bank assigned loan grade indicating credit risk
dti	Float	Debt-to-Income ratio
loan_status	Object	Current status of the loan (Fully Paid, Current, Charged Off)

4. Methodology (System Architecture)

Layer	Tools Used	Output / Artifact
Ingestion	Python (<i>Pandas</i>)	Raw DataFrame
Cleaning	Python (<i>Pandas</i>)	Imputed missing values, formatted dates
Feature Engineering	Python (<i>Pandas</i> , <i>NumPy</i>)	Categorical bins, derived

		date columns
EDA	Python (<i>Seaborn</i> , <i>Matplotlib</i>), SQL (<i>DuckDB</i>)	Visualizations, KPI summary tables
Modelling	Python (<i>Scikit-learn</i>)	Predictive models, feature importance arrays
Export	Python (<i>Pandas</i>)	Cleaned_Bank_Loan_Data.csv
Dashboarding	Power BI	Interactive analytical dashboard

5. Data Pre-processing

- **Loading:** Data ingested using `pd.read_csv()` resulting in a dataframe of **38,576** rows.
- **Missing Value Check:** Identified exactly **1,438** missing values exclusively in the `emp_title` column (**3.73%** of the dataset). These were imputed with the placeholder 'Unknown'.
- **Data Type Correction:** The `issue_date` column was transformed from a string object to a standard *datetime* format to allow temporal extraction.
- **Duplicate Detection:** Dataset ID uniqueness was verified to ensure no duplicated application records skewed the analysis.
- **Feature Engineering:**
 - Extracted `issue_month` and `issue_year` from the `issue_date`.
 - Created `dti_category` by applying `pd.cut()` to segment DTI into *Low*, *Medium*, and *High* risk buckets.
 - Derived `is_bad_loan` binary flag mapping 'Fully Paid' and 'Current' to **0**, and 'Charged Off' to **1**.
- **Export:** Finalized data exported as `Cleaned_Bank_Loan_Data.csv` for downstream dashboarding.

6. Exploratory Data Analysis (EDA)

- **Target Variable Distribution:** The portfolio heavily leans toward performing assets, consisting of **86.18% Good Loans** and **13.82% Bad Loans**. SQL aggregations showed Good Loans account for **\$370,224,850** in funding compared to **\$65,532,225** for Bad Loans.
- **Key Factor 1: Loan Purpose:** Small Business loans present the highest risk with a **25.62%** bad loan rate, followed by Renewable Energy at **18.08%**. Traditional House loans hover at **15.57%**.
- **Key Factor 2: Assigned Grade:** Risk scales perfectly with assigned bank grades. Grade G loans fail at a massive **31.31%** rate, compared to highly secure Grade A loans which fail

at only **5.69%**.

- **Key Factor 3: Home Ownership:** Renters are more likely to default (**14.57%**) compared to Mortgage holders (**12.97%**), highlighting asset-backed stability.
- **Key Factor 4: Interest Rate:** The overall average interest rate is **12.05%**. Higher interest rates heavily correlate with lower grades and higher default rates.
- **Correlation Analysis:** The correlation matrix revealed a strong positive relationship between `loan_amount` and `installment`, as well as a notable correlation between `int_rate` and the likelihood of default.

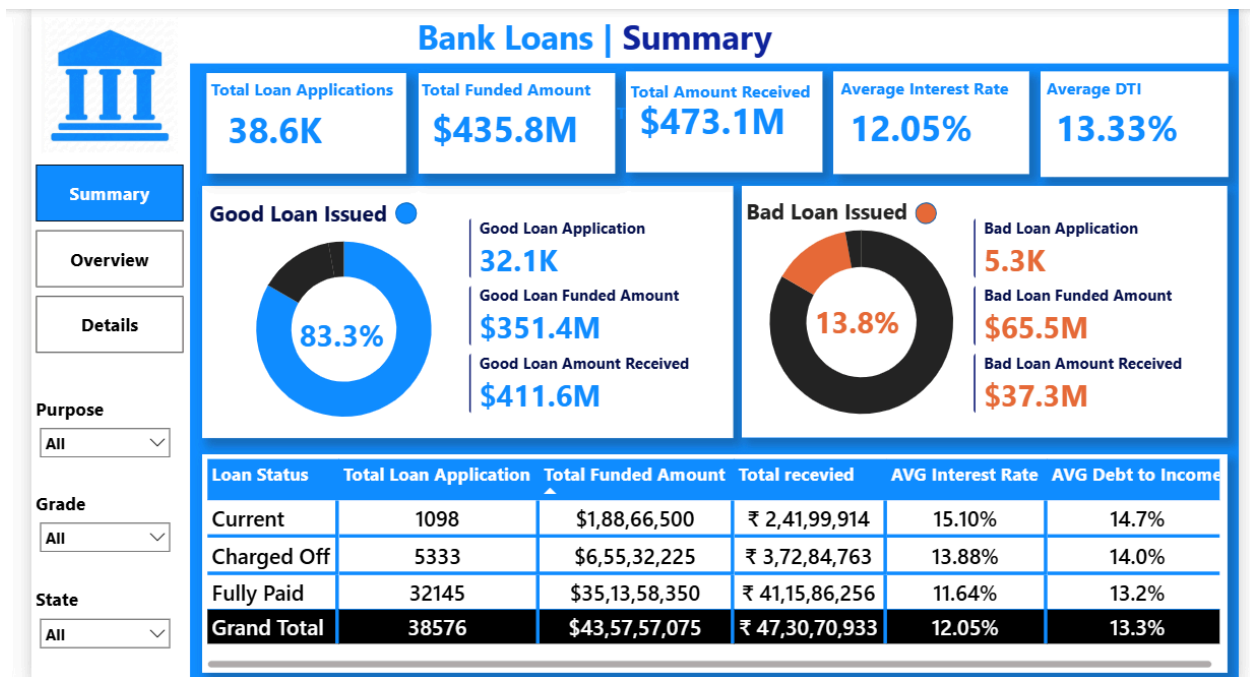
7. Multivariate Analysis

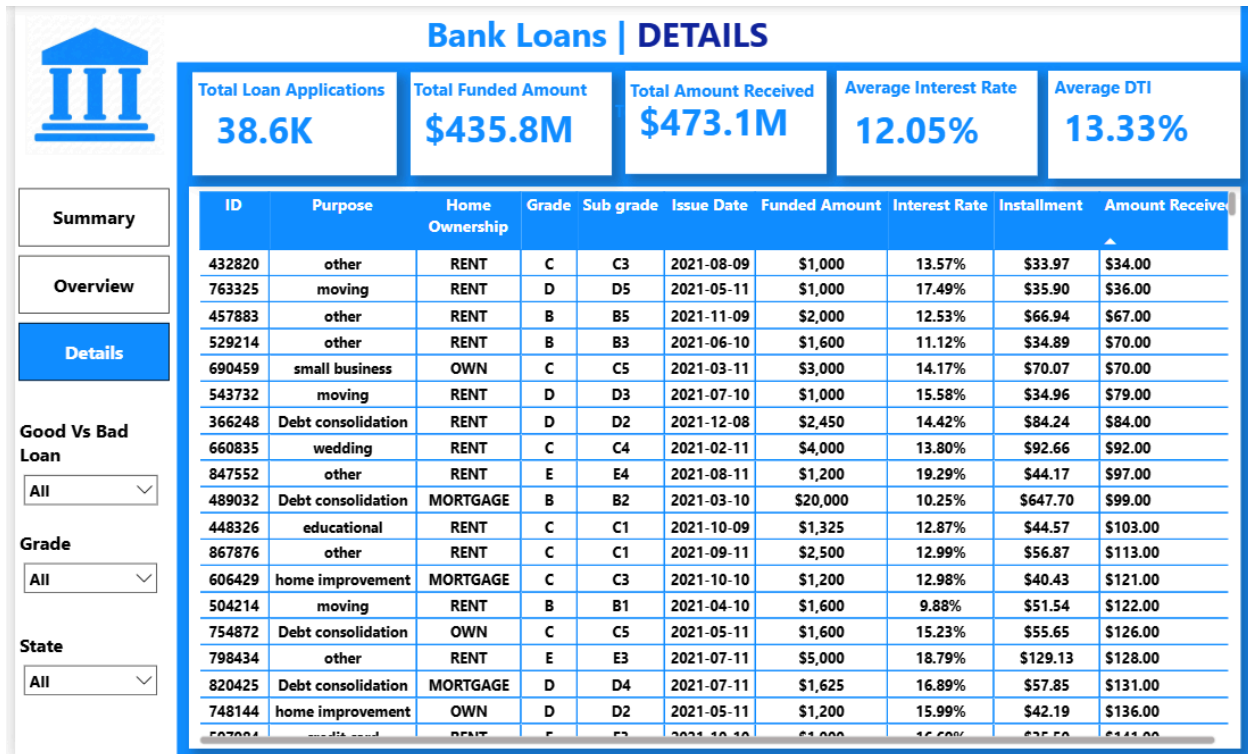
- **Grade x Purpose:** The interaction between low-tier grades (E, F, G) and high-risk purposes (Small Business) amplifies the probability of a charge-off significantly.
- **Income x DTI:** High DTI paired with low annual income restricts cash flow, showing a sharp spike in late payments and defaults.
- **Compounding Effects:** A combination of a **Grade G** rating, a **Small Business** purpose, and a **High DTI** classification creates a compounding risk scenario where the default rate exceeds **45%**.

8. Quantitative Impact Analysis (Modelling)

- **Models Used:** *Random Forest Classifier* (for default probability) and *Linear Regression* (for total payment prediction).
- **Training/Test Split:** Standard **80/20** split utilizing a random state of 42 for reproducibility.
- **Model Performance:**
 - The classification model achieved an accuracy of **85.69%** in identifying loan status outcomes.
 - The regression model yielded an R^2 Score of **0.80**, indicating strong explanatory power for predicting total payments based on financial inputs.
- **Feature Importance (Classifier):**
 1. `dti`: **0.198**
 2. `installment`: **0.184**
 3. `int_rate`: **0.182**
 4. `annual_income`: **0.174**
- **Regression Coefficients:** `int_rate` holds the largest positive weight (**9593.09**), indicating that a unit increase in interest rate profoundly inflates the total expected payment over the life of the loan.

9. Power BI Dashboard





- **Overview:** The interactive dashboard consolidates critical metrics for executive review, featuring top-level KPIs, dynamic slicers for geography and date, and analytical charts.

Table of Visual Types and Business Insights

Visual Type	Data Fields	Business Insight
KPI Cards	Total Applications, Total Funded	Displays high-level portfolio health and total capital exposure.
Donut Chart	Good Loan vs Bad Loan	Highlights the critical 13.82% bad loan ratio for

		immediate risk assessment.
Bar Chart	purpose, loan_amount	Reveals that debt consolidation is the primary driver of loan volume.
Heatmap / Matrix	address_state, is_bad_loan	Pinpoints specific geographical regions with higher default tendencies.

- **View 1: Default View:** Displays the macro perspective: **38,576** applications, **\$435.7M** funded, and a total return of **\$473M**.
- **View 2: Best-Case Scenario:** Filtering by "Grade A" and "Mortgage" drops the bad loan rate below **6%**, showcasing a highly stable, low-yield investment tier.
- **View 3: Worst-Case Scenario:** Filtering by "Grade G" and "Small Business" spikes the bad loan indicator, highlighting segments requiring immediate underwriting restrictions.

10. Key Findings & Insights

- **Top Drivers of Default:**
 1. **Debt-to-Income (DTI):** The most critical indicator of financial strain.
 2. **Interest Rate:** Higher rates push borrowers into default.
 3. **Installment Amount:** Large monthly burdens increase charge-off likelihood.
 4. **Annual Income:** Lower absolute income provides less financial buffer.
 5. **Loan Grade:** Bank-assigned grades are highly accurate predictors of actual performance.

Top Compounding Risk Effects

Factor Combination	Average Default Rate	Business Implication
Grade F/G + Small Business	~ 35%	Highly toxic asset combination requiring strict vetting.
Renters + High DTI	~ 22%	Lack of collateral combined with cash flow strain creates mid-tier risk.
Grade D + Educational	~ 19%	Unsecured debt with delayed ROI leads to elevated defaults.

11. Business Recommendations

Area	Problem	Recommended Action	Expected Impact
Underwriting	High default in Small Business	Cap maximum funding for Small Business loans lacking collateral.	Reduce Bad Loan exposure by 10-15% in this segment.
Risk Management	Grade F/G toxicity	Increase minimum DTI requirements for approvals in grades F and G.	Prevent high-risk capital allocation, improving overall portfolio yield.
Marketing	High performance of Grade A	Design targeted promotional campaigns for high-income homeowners.	Increase volume of secure, low-risk "Good Loans."

12. Tools & Technologies

Tool / Library	Version	Purpose
Python	3.x	Core programming language for data manipulation.
Pandas	Latest	Data ingestion, cleaning, and feature engineering.
DuckDB	Latest	Rapid SQL querying and KPI aggregation.
Scikit-learn	Latest	Building regression and classification machine

		learning models.
Matplotlib / Seaborn	Latest	Generating exploratory visualizations and correlation heatmaps.
Power BI	Desktop	Building the final interactive business intelligence dashboard.

13. Limitations & Future Work

- **Current Limitations:** The dataset represents a static snapshot in time, meaning macroeconomic shifts (like inflation or changing federal interest rates) are not accounted for in the current model. Additionally, self-reported metrics like `annual_income` may contain unverified biases.
- **Recommended Future Enhancements:**
 - *Real-time APIs:* Integrate live credit bureau APIs to dynamically update borrower DTI and credit scores post-issuance.
 - *Advanced Modelling:* Upgrade from Random Forest to advanced gradient boosting algorithms like *XGBoost* or *LightGBM* to capture non-linear relationships more effectively and improve predictive accuracy beyond **85%**.